

Securd — Asset Listing & Risk Parameters

DeFi Risk Report | XRPL EVM Mainnet (Chain ID 1440000) & XRP Ledger Mainnet

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Executive Summary

Category	Available Today	Future
Single assets	XRP, USDC, WETH, WBTC, USDT, mXRP	RLUSD
LP tokens	XRP/RLUSD, XRP/USDC, XRP/ARMY — Axelar ITS registration in progress	—

WBTC (\$241 total supply) and USDT (\$207 total supply) are deployed on XRPL EVM but require seeding before activation as meaningful markets.

Axelar is actively working on registering XRPL native AMM LP tokens in ITS to make them bridgeable to XRPL EVM. Parameters are defined and ready for deployment upon registration completion.

Risk Parameter Definitions

Parameter	Description
Collateral Factor (CF)	Maximum percentage of collateral value that can be borrowed against. A CF of 75% means a user depositing \$100 can borrow up to \$75. Liquidation is triggered as soon as the borrowed value exceeds CF × collateral value.
Close Factor	Maximum percentage of a borrower's outstanding debt that a liquidator can repay in a single transaction. Set at 50% for all markets to prevent full liquidation in one block.
Reserve Factor	Percentage of borrower interest redirected to the protocol insurance reserve instead of suppliers. At 20%, if borrowers pay 10% APY, suppliers receive 8% APY and 2% accrues to the reserve.
Supply Cap	Maximum total amount of an asset that can be deposited into the market. Limits protocol exposure to any single asset.
Borrow Cap	Maximum total amount that can be borrowed from the market. Set at 80% of supply cap to guarantee a minimum 20% liquidity buffer for supplier withdrawals at all times.
Liquidation Incentive (LI)	Bonus paid to liquidators expressed as a percentage of the debt repaid. A 10% LI means a liquidator repaying \$100 of debt receives \$110 worth of collateral.
IR — Base Rate	Minimum annual borrow rate regardless of utilization.
IR — Slope 1	Rate of interest increase per unit of utilization below the kink (normal regime).
IR — Slope 2	Rate of interest increase per unit of utilization above the kink (stress regime). Intentionally steep to force repayment and attract new supply when the market approaches full utilization.
IR — Kink	Optimal utilization target. Below this level interest rates are low and stable; above it rates rise sharply to protect liquidity.

Health Factor (HF):

$$HF = \frac{\sum(\text{collateral}_i \times CF_i \times \text{price}_i)}{\sum(\text{borrow}_j \times \text{price}_j)}$$

Liquidation is triggered when **HF < 1.0**. Maximum borrowing is reached when HF = 1.0.

PART 1 — SINGLE ASSETS AVAILABLE TODAY

XRP

What it is: Native gas token of XRPL EVM. 6th largest crypto asset globally, with the deepest on-chain liquidity of any asset in this ecosystem.

Market data (live):

- Price: **\$1.39** | Supply: Native | Liquidity: Deepest on-chain — all AMMs are XRP-denominated

Role: Collateral + borrowable

Risk profile:

- Volatility: High. XRP has experienced 80-90% drawdowns in bear cycles.
- Liquidation note: For positions where HF drops below 0.825, partial liquidations reduce borrower equity faster than debt is relieved. The protocol bad debt insurance reserve absorbs residual exposure.
- Oracle: XRP/RLUSD AMM TWAP (primary) + Bitstamp DEX order book (secondary). Max staleness: 200 ledgers.
- Smart contract risk: None — native asset.

Parameter	Value
Collateral Factor	75%
Close Factor	50%
Reserve Factor	20%
Supply Cap	3,500,000 XRP
Borrow Cap	2,800,000 XRP
Liquidation Incentive	10%
IR — Base Rate	0%
IR — Slope 1	5%/year
IR — Slope 2	85%/year
IR — Kink	80%

Borrow APY at utilization:

- 50% → 2.5% | 80% → 4.0% (kink) | 95% → 16.75%

USDC (Axelar)

EVM address: 0xa16148c6Ac9EDe0D82f0c52899e22a575284f131

XRPL issuer (Axelar gateway): rfmS3zqrQrka8wVyhXifEeyTwe8AMz2Yhw

Market data (live):

- Price: ~\$1.00 | 4,153 USDC on XRPL EVM | \$956K AMM TVL | \$129K 7d volume

Role: Collateral + borrowable

Risk profile:

- Stability: Excellent — regulated stablecoin, monthly reserve attestations by Circle.
- Bridge risk: Axelar-wrapped. If the bridge is compromised, USDC becomes unbacked.
- Oracle: XRP/USDC AMM TWAP × XRP/USD. Circuit breaker if price < \$0.97 or > \$1.05. Max staleness: 200 ledgers.

Parameter	Value
Collateral Factor	80%
Close Factor	50%
Reserve Factor	20%
Supply Cap	50,000 USDC
Borrow Cap	40,000 USDC
Liquidation Incentive	5%
IR — Base Rate	0%
IR — Slope 1	8%/year
IR — Slope 2	75%/year
IR — Kink	90%

Borrow APY at utilization:

- 50% → 4.0% | 90% → 7.2% (kink) | 95% → 10.95% | 100% → 14.7%

USDT (Axelar)

EVM address: 0x9F8CF9c00fac501b3965872f4ed3271f6f4d06fF

XRPL issuer (Axelar gateway): rfmS3zqrQrka8wVyhXifEeyTwe8AMz2Yhw

Market data (live):

- Price: ~\$1.00 | **207 USDT** on XRPL EVM | \$217K AMM TVL | \$25K 7d volume

Role: Collateral + borrowable

Risk profile:

- Stability: Good — most widely used stablecoin globally; reserve composition less transparent than Circle.
- Bridge risk: Same Axelar wrapping as USDC.
- Oracle: XRP/USDT AMM TWAP $\times$ XRP/USD. TWAP window extended to 50 ledgers given thinner AMM depth. Max staleness: 200 ledgers.
- **Minimum seeding required before activation: \$10,000 USDT.** Activating on an empty market exposes the protocol to initial exchange rate manipulation.

Parameter	Value
Collateral Factor	78%
Close Factor	50%
Reserve Factor	20%
Supply Cap	50,000 USDT
Borrow Cap	40,000 USDT
Liquidation Incentive	5%
IR — Base Rate	0%
IR — Slope 1	8%/year
IR — Slope 2	75%/year
IR — Kink	90%

Borrow APY at utilization:

- 50% $\rightarrow$ 4.0% | 90% $\rightarrow$ 7.2% (kink) | 95% $\rightarrow$ 10.95% | 100% $\rightarrow$ 14.7%

WETH (Axelar)**EVM address:** 0x50498dC52bCd3dAeB54B7225A7d2FA8D536F313E**XRPL issuer (Axelar gateway):** rfmS3zqrQrka8wVyhXifEeyTwe8AMz2Yhw**Market data (live):**

- Price: ~\$2,127 (1,530 XRP/WETH $\times$ \$1.39) | **2.577 WETH** (~\$5,482) on XRPL EVM | 104.7 WETH bridged on XRPL

Role: Collateral + borrowable**Risk profile:**

- Volatility: High. ETH has experienced 80%+ drawdowns in bear markets. Positions below HF = 0.825 generate bad debt under partial liquidation.

- Bridge risk: Axelar wrapping — same risk profile as USDC.
- Oracle: WETH/XRP XRPL DEX TWAP × XRP/USD. Falls back to XRPL ledger DEX if WETH EVM DEX liquidity < \$10K. Max staleness: 200 ledgers.
- **Minimum seeding: 1 WETH before activating.**

Parameter	Value
Collateral Factor	75%
Close Factor	50%
Reserve Factor	20%
Supply Cap	20 WETH
Borrow Cap	16 WETH
Liquidation Incentive	10%
IR — Base Rate	0%
IR — Slope 1	5%/year
IR — Slope 2	82%/year
IR — Kink	85%

Borrow APY at utilization:

- 50% → 2.5% | 85% → 4.25% (kink) | 95% → 12.45%

Supply cap roadmap: 20 → 50 → 100 WETH. Each step requires 2 weeks of stable operation at the prior level.

WBTC (Axelar)

EVM address: 0xF8Eb4Ed0d4CF2bb707c0272F8C6827dEB6e4C0A9

XRPL issuer (Axelar gateway): rfmS3zqrQrka8wVyhXi fEeyTwe8AMz2Yhw

Market data (live):

- Price: ~\$105,000 | **0.002 WBTC** (~\$241) on XRPL EVM | 0.0067 BTC bridged on XRPL

Role: Collateral + borrowable (once seeded)

Risk profile:

- Volatility: High. BTC experiences 70–80% drawdowns in bear cycles.
- Bridge risk: Axelar wrapping adds custodial risk on top of the underlying asset.
- Oracle: WBTC/XRP on the **XRPL ledger DEX** × XRP/USD — XRPL EVM DEX has near-zero WBTC liquidity and cannot serve as a reliable price source. Collateral use remains disabled until XRPL EVM

DEX liquidity exceeds \$50,000.

- **Minimum seeding: 0.1 WBTC before activating.**

Parameter	Value
Collateral Factor	73% (disabled for collateral use until DEX liquidity > \$50K)
Close Factor	50%
Reserve Factor	20%
Supply Cap	0.75 WBTC
Borrow Cap	0.60 WBTC
Liquidation Incentive	10%
IR — Base Rate	0%
IR — Slope 1	5%/year
IR — Slope 2	82%/year
IR — Kink	80%

Borrow APY at utilization:

- 50% → 2.5% | 80% → 4.0% (kink) | 95% → 16.3%

mXRP (Midas)

EVM address: `0x06e0B0F1A644Bb9881f675Ef266CeC15a63a3d47`

Market data (live):

- Price: ~\$1.44 (1.035 XRP × \$1.39) | **3,756,955 mXRP** (~\$5.4M) on XRPL EVM | 2,384 XRPL holders

Role: Collateral only — no borrowing

Risk profile:

- Midas protocol risk: If Midas is exploited or becomes insolvent, mXRP depegs from XRP. The 3.5% premium disappears instantly under stress.
- Double risk layer: Midas smart contract + Axelar bridge — two independent failure points.
- Oracle: mXRP/XRP DEX TWAP × XRP/USD. Price is hard-capped at `min(live_price, 1.06 × XRP_USD)` to prevent manipulation via premium inflation.
- Borrow cap set to zero — circular leverage must be prevented.

Parameter	Value
Collateral Factor	60%
Close Factor	50%
Reserve Factor	20%
Supply Cap	5,500,000 mXRP
Borrow Cap	0 — supply only
Liquidation Incentive	10%
IR — Base Rate	N/A — no borrowing market
IR — Slope 1	N/A
IR — Slope 2	N/A
IR — Kink	N/A

Monitoring triggers: $mXRP/XRP < 0.98 \rightarrow$ reduce CF to 50%. $mXRP/XRP < 0.95 \rightarrow$ pause market.

PART 2 — LP TOKENS (AXELAR ITS REGISTRATION IN PROGRESS)

Axelar is actively working on registering XRPL native AMM LP tokens in its Interchain Token Service (ITS), which will make them bridgeable to XRPL EVM as standard ERC-20 tokens. The risk parameters below are defined and approved — markets will be deployed on Securd as soon as each LP token's Axelar ITS registration is confirmed.

LP tokens are accepted as **collateral only**. They cannot be borrowed, and suppliers do not earn interest on deposited LP tokens. Their sole purpose is to unlock borrowing capacity against an existing LP position. Accordingly, no interest rate model applies to these markets and the reserve factor is 0% as no interest income is generated.

Technical requirements upon Axelar registration:

1. LP token registered in Axelar ITS with MINT_BURN token manager
2. AMM account on XRPL authorizes Axelar to issue the LP token cross-chain
3. Lending market deployed on XRPL EVM for the bridged LP token
4. Oracle contract deployed with validation guard: `require(lp_supply > 0 && xrp_reserve > 0 && stable_reserve > 0)`

XRP/RLUSD LP — Priority 1

Pool data	Value
XRPL AMM account	rhWTXC2m2gGGA9WozUaoMm6kLAVPb1tcS3
LP token currency (XRPL)	037C2A57B0011520DE389E332043EC0FAF858ACE
LP token issuer (XRPL)	rhWTXC2m2gGGA9WozUaoMm6kLAVPb1tcS3
XRP reserve	1,893,541 XRP
RLUSD reserve	2,499,660 RLUSD
LP token supply	3,165,547,749
TVL	\$5.26M
7d volume	\$636K
Pool fee	0.267%
LP annual yield	2.9%

Role: Collateral only — cannot be borrowed, no interest earned on deposit

Oracle formula (geometric mean — manipulation resistant):

$$LP_price_USD = 2 \times \sqrt{XRP_reserve \times P_xrp \times RLUUSD_reserve \times 1.00} / LP_supply$$

Current LP unit price: $2 \times \sqrt{1,893,541 \times \$1.39 \times 2,499,660} / 3,165,547,749 \approx \0.00166

Parameter	Value
Collateral Factor	62%
Close Factor	50%
Reserve Factor	0% — collateral-only market, no interest income
Supply Cap	\$526,000 worth of LP tokens
Borrow Cap	0 — collateral only
Liquidation Incentive	10%
IR — Base Rate	N/A — no borrowing market
IR — Slope 1	N/A
IR — Slope 2	N/A
IR — Kink	N/A

XRP/USDC LP — Priority 2

Pool data	Value
XRPL AMM account	rM7cHVPfhe9yxQNk2kDNBEQqoQmMcQGPWE
LP token currency (XRPL)	03USDC_LP (resolved on Axelar ITS registration)
LP token issuer (XRPL)	rM7cHVPfhe9yxQNk2kDNBEQqoQmMcQGPWE
XRP reserve	343,875 XRP
USDC reserve	343,875 USDC (approx.)
TVL	\$956K
7d volume	\$129K
Pool fee	0.202%
LP annual yield	2.7%

Role: Collateral only — cannot be borrowed, no interest earned on deposit

Oracle formula: $LP_price_USD = 2 \times \sqrt{XRP_reserve \times P_xrp \times USDC_reserve \times 1.00} / LP_supply$

Parameter	Value
Collateral Factor	57%
Close Factor	50%
Reserve Factor	0% — collateral-only market, no interest income
Supply Cap	\$95,600 worth of LP tokens
Borrow Cap	0 — collateral only
Liquidation Incentive	10%
IR — Base Rate	N/A — no borrowing market
IR — Slope 1	N/A
IR — Slope 2	N/A
IR — Kink	N/A

XRP/ARMY LP — Priority 3 (conditional on external price feed)

Additional condition: ARMY token must have an external price feed independent of this pool (CoinGecko listing or centralized exchange) before this market can be activated.

Pool data	Value
XRPL AMM account	rnsRq5ahgbFeRiAgBVvFTafyAgiS9x9Ztn
LP token currency (XRPL)	037C2A57B0011520DE389E332043EC0FAF858ACE (ARMY pool LP currency — different from RLUSD pool)
LP token issuer (XRPL)	rnsRq5ahgbFeRiAgBVvFTafyAgiS9x9Ztn
ARMY token issuer	rGG3wQ4kUzd7Jnmk1n5NWPZjjut62kCBfC
XRP reserve	244,601 XRP
ARMY reserve	41,500,094 ARMY
LP token supply	3,165,547,749
TVL	~\$680K
7d volume	Low (0.001% fee generates minimal volume signal)
Pool fee	0.001%
LP annual yield	~0%

Role: Collateral only — cannot be borrowed, no interest earned on deposit

Oracle: $LP_price_USD = 2 \times XRP_reserve \times P_xrp / LP_supply$

ARMY is priced at \$0 — only the XRP side of the pool is credited. This is conservative by design: if ARMY holds any market value, the actual collateral exceeds Securd's credit.

Current LP unit price (XRP-only): $2 \times 244,601 \times \$1.39 / 3,165,547,749 \approx \0.000215

Parameter	Value
Collateral Factor	35%
Close Factor	50%
Reserve Factor	0% — collateral-only market, no interest income
Supply Cap	\$34,000 worth of LP tokens
Borrow Cap	0 — collateral only
Liquidation Incentive	15%
IR — Base Rate	N/A — no borrowing market
IR — Slope 1	N/A
IR — Slope 2	N/A
IR — Kink	N/A

PART 3 — FUTURE LISTINGS

Asset not yet available on XRPL EVM mainnet.

RLUSD

What it is: Ripple's regulated USD stablecoin. NYDFS-licensed, backed 1:1 by USD deposits and US T-Bills. Deepest liquidity on XRPL (\$5.26M in XRP/RLUSD AMM, \$636K 7d volume).

Current status: XRPL-native only. Not yet on XRPL EVM.

Listing trigger: RLUSD bridged to XRPL EVM with > \$100,000 supply.

Parameter	Value
Collateral Factor	83%
Close Factor	50%
Reserve Factor	20% → reduce to 10% once insurance fund > \$100K
Supply Cap	2,000,000 RLUSD
Borrow Cap	1,600,000 RLUSD
Liquidation Incentive	5%
IR — Base Rate	0%
IR — Slope 1	8%/year
IR — Slope 2	75%/year
IR — Kink	90%

Borrow APY at utilization:

- 50% → 4.0% | 90% → 7.2% (kink) | 95% → 10.95% | 100% → 14.7%

PART 4 — COMPLETE PARAMETER SUMMARY

Single asset markets

Asset	CF	Close Factor	Liq. Incentive	Reserve Factor	Supply Cap	Borrow Cap	IR Kink
XRP	75%	50%	10%	20%	3,500,000 XRP	2,800,000 XRP	80%
USDC	80%	50%	5%	20%	50,000 USDC	40,000 USDC	90%
USDT	78%	50%	5%	20%	50,000 USDT	40,000 USDT	90%
WETH	75%	50%	10%	20%	20 WETH	16 WETH	85%
WBTC †	73%	50%	10%	20%	0.75 WBTC	0.60 WBTC	80%
mXRP	60%	50%	10%	20%	5,500,000 mXRP	0	N/A

† WBTC collateral use disabled until XRPL EVM DEX liquidity > \$50K

LP token markets (Axelar ITS registration in progress)

Asset	CF	Close Factor	Liq. Incentive	Reserve Factor	Supply Cap	Borrow Cap
XRP/ RLUSD LP	62%	50%	10%	0%	\$526,000	0
XRP/USDC LP	57%	50%	10%	0%	\$95,600	0
XRP/ARMY LP ‡	35%	50%	15%	0%	\$34,000	0

‡ XRP/ARMY LP requires external ARMY price feed before activation

Future markets

Asset	CF	Liq. Incentive	Reserve Factor	Listing trigger
RLUSD	83%	5%	20%	Bridged to XRPL EVM > \$100K supply

PART 5 — INTEREST RATE MODELS

Interest rates follow a two-slope model with a utilization kink. Below the kink, rates rise gradually to balance supply and demand. Above the kink, rates accelerate sharply to incentivize new supply and encourage borrower repayment before the market reaches full utilization.

Formula:

$$\begin{aligned} \text{borrowRate}(u) &= \text{Slope1} \times u && \text{when } u \leq \text{kink} \\ \text{borrowRate}(u) &= \text{Slope1} \times \text{kink} + \text{Slope2} \times (u - \text{kink}) && \text{when } u > \text{kink} \\ \text{supplyRate}(u) &= \text{borrowRate}(u) \times (1 - \text{reserveFactor}) \times u \end{aligned}$$

Asset	Slope 1	Slope 2	Kink	Rate at kink	Rate at 100%
XRP	5%/yr	85%/yr	80%	4.00%	17.75%
USDC	8%/yr	75%/yr	90%	7.20%	14.70%
USDT	8%/yr	75%/yr	90%	7.20%	14.70%
WETH	5%/yr	82%/yr	85%	4.25%	16.05%
WBTC	5%/yr	82%/yr	80%	4.00%	16.40%
RLUSD	8%/yr	75%/yr	90%	7.20%	14.70%
mXRP	N/A	N/A	N/A	N/A	N/A
LP tokens	N/A	N/A	N/A	N/A	N/A

PART 6 — ORACLE ARCHITECTURE

Reliable price feeds are the most critical component of any lending protocol. An incorrect or manipulated price can allow under-collateralized borrowing or prevent valid liquidations. Securd uses on-chain price sources with time-weighted averaging and multiple protective layers.

How prices are computed

Native XRP

XRP price is derived from two independent on-chain sources and cross-checked against each other:

1. **Primary:** Time-weighted average of the XRP/RLUSD AMM pool reserve ratio over the last 20 XRPL ledgers (~20 seconds). The \$5.26M TVL of this pool makes large price manipulation extremely expensive.
2. **Secondary:** Bitstamp USD/XRP order book on the XRPL native DEX, used as a sanity check. If the two sources diverge by more than 3%, the oracle raises an alert.

Stablecoins (USDC, USDT, RLUSD)

Stablecoin prices are treated as \$1.00 by default. The oracle continuously monitors the on-chain AMM ratio:

- USDC: derived from the XRP/USDC AMM pool (TVL \$956K) — $P_{usdc} = P_{xrp} / (XRP_reserve / USDC_reserve)$

- USDT: derived from the XRP/USDT AMM pool (TVL \$217K) — same ratio formula, 50-ledger TWAP to compensate for thinner liquidity

- RLUSD: fixed at \$1.00 with AMM ratio as secondary validation

If the derived stablecoin price deviates from \$1.00 by more than 3% in either direction, the circuit breaker activates.

Wrapped assets (WETH, WBTC)

Prices are computed in two steps:

1. Derive the asset/XRP ratio from the XRPL DEX TWAP (20 ledgers)
2. Multiply by the XRP/USD price

$$P_{weth} = (WETH/XRP \text{ DEX TWAP}) \times P_{xrp}$$

WBTC uses the XRPL ledger DEX (not the XRPL EVM DEX, which has near-zero WBTC liquidity). Collateral use is disabled until XRPL EVM DEX liquidity exceeds \$50K.

mXRP

$$P_{mxrp} = \min(mXRP/XRP \text{ DEX TWAP} \times P_{xrp}, 1.06 \times P_{xrp})$$

The hard cap at 1.06× prevents an attacker from pumping the mXRP/XRP premium to extract inflated borrowing power. The cap is reviewed quarterly against the observed trading premium.

LP tokens

LP token value is derived using the geometric mean of pool reserves, which is resistant to single-block manipulation:

XRP/Stablecoin pairs:

$$\text{LP_price_USD} = 2 \times \sqrt{\text{XRP_reserve} \times P_{\text{xrp}} \times \text{Stable_reserve} \times 1.00} / \text{LP_supply}$$

XRP/ARMY (conservative):

$$\text{LP_price_USD} = 2 \times \text{XRP_reserve} \times P_{\text{xrp}} / \text{LP_supply}$$

ARMY is valued at \$0 — the LP price reflects only the XRP side. This is deliberately conservative: if ARMY holds any market value, the actual collateral is worth more than Securd credits.

Oracle guard (required in contract):

```
require(lp_supply > 0 && xrp_reserve > 0 && stable_reserve > 0, "invalid pool state");
```

TWAP parameters

Asset	TWAP window	Rationale
XRP	20 ledgers	Deep liquidity (\$5.26M AMM) — short window sufficient
USDC	20 ledgers	Deep liquidity (\$956K AMM)
USDT	50 ledgers	Thinner liquidity (\$217K AMM) — wider window reduces manipulation risk
WETH	20 ledgers	XRPL DEX volume is sufficient
WBTC	20 ledgers	XRPL ledger DEX is used
mXRP	20 ledgers	mXRP/XRP premium is stable; capped independently
LP tokens	20 ledgers	Pool reserves sampled per ledger; geometric mean absorbs outliers

Staleness protection

If an oracle has not been updated within the defined window, the protocol escalates through three stages:

Stage	Condition	Action
Stale	Last update > 200 ledgers	Block new borrows using this price; existing positions unaffected
Critical	Last update > 1,000 ledgers (~17 min)	Full market pause — no borrows, no new collateral deposits
Recovery	Price feed restored	Manual admin review required before un-pausing

Circuit breakers

Asset	Trigger	Action
XRP	Price moves > 20% in 1 hour	Pause new borrows; alert governance
USDC / USDT / RLUSD	Price < \$0.97 or > \$1.05	Auto-pause market
WETH	Price moves > 15% in 1 hour	Pause new borrows
WBTC	Price moves > 15% in 1 hour	Pause new borrows
WBTC	XRPL EVM DEX liquidity < \$50K	CF = 0; deposit and borrow allowed, no collateral use
mXRP	mXRP/XRP ratio < 0.98	CF reduced to 50%
mXRP	mXRP/XRP ratio < 0.95	Full market pause
LP tokens	XRP reserve changes > 15% in one ledger	Pause LP oracle and market

PART 7 — LIQUIDATION MECHANICS & BAD DEBT

Liquidation flow

When a borrower's Health Factor drops below 1.0, any address can trigger a liquidation:

1. Liquidator selects a borrower with $HF < 1.0$ and a market to repay
2. Liquidator repays up to 50% of the borrower's outstanding debt in that market (close factor)
3. Liquidator receives collateral worth $\text{debt_repaid} \times (1 + \text{liquidation_incentive})$ at the oracle price
4. Borrower's HF rises; if still < 1.0 , further liquidations are possible

Bad debt threshold

In this protocol's liquidation model, partial liquidation only improves the borrower's HF when $HF > CF \times (1 + LI)$. Below this threshold, each partial liquidation reduces the borrower's collateral faster than it reduces debt, and the position generates bad debt. The protocol insurance reserve (funded by the reserve factor) absorbs this shortfall.

Asset	CF	Liq. Incentive	Bad debt if HF below
XRP	75%	10%	0.825
USDC	80%	5%	0.840
USDT	78%	5%	0.819
WETH	75%	10%	0.825
WBTC	73%	10%	0.803
mXRP	60%	10%	0.660
XRP/RLUSD LP	62%	10%	0.682
XRP/USDC LP	57%	10%	0.627
XRP/ARMY LP	35%	15%	0.402

The protocol must maintain a **bad debt insurance reserve of at least 3% of total protocol TVL**, accumulated continuously from reserve factor income.

All critical parameter changes (CF, supply caps, liquidation incentive) are subject to a **48-hour governance timelock**.

PART 8 — RISK WATCHLIST

Risk	Affected assets	Trigger	Action
Axelar bridge exploit	USDC, USDT, WETH, WBTC, mXRP	Bridge TVL drops > 20% in 1h	Pause all Axelar-dependent markets
mXRP depeg	mXRP	mXRP/XRP < 0.98	Reduce CF to 50%
mXRP depeg — severe	mXRP	mXRP/XRP < 0.95	Pause market
Stablecoin depeg	USDC, USDT	Price < \$0.97	Auto-pause; manual review required
Oracle staleness	Any	Last update > 200 ledgers	Block new borrows
Oracle staleness — severe	Any	Last update > 1,000 ledgers	Full market pause
WBTC oracle invalid	WBTC	XRPL EVM DEX liquidity < \$50K	CF = 0; deposit/borrow only
Bad debt accumulation	Any	Position HF < CF × (1 + LI)	Reserve fund absorption; socialization if > 2% TVL
XRP flash crash	XRP, all LP tokens	XRP drops > 30% in 1 ledger	Circuit breaker; pause new borrows
LP pool drain	LP tokens	XRP reserve moves > 15% in 1 ledger	Pause LP oracle and market
Governance attack	All	Admin proposes CF reduction	48h timelock enforced before execution
Empty market seeding risk	USDT, WBTC	First deposit on near-empty market	Minimum \$10K seed required before activation